

Python Programming

Unit 01 – Lecture 01 Notes

Introduction and Working with Python

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1 Lecture Overview

This lecture helps you **start** with Python the right way:

- what Python is and why it is popular,
- how to run Python in **interactive mode (REPL)** and **scripting mode**,
- and how to write a first program that follows the common pattern: **Input → Process → Output**.

Repository (course materials): <https://github.com/tali7c/Python-Programming>

2 Core Concepts

2.1 What is Python? (Quick Picture)

Python is a general-purpose programming language known for:

- **readable syntax** (you can focus on logic, not symbols),
- a **large ecosystem** of libraries (web, data analysis, AI, automation),
- and **cross-platform** support.

In this course, Python is used to build programming fundamentals that later support subjects like data structures, databases, and data analysis.

2.2 Two Ways to Run Python

There are two everyday ways to run Python code:

1. **Interactive mode (REPL):** You type a statement and get the result immediately. Good for exploration.
2. **Scripting mode:** You save a program in a `.py` file and execute it. Good for assignments, labs, and reusable programs.

2.2.1 Interactive Mode (REPL)

Example (Python prompt):

```
>>> 2 + 3
5
>>> "UPES".lower()
'upes'
```

Use REPL when you want quick feedback, or when you are learning a new concept.

2.2.2 Scripting Mode (`.py` file)

A script is a plain text file that contains Python statements. Typical run command in a terminal:

```
python my_program.py
```

If your file is `task.py`, the command is `python task.py`.

2.3 The IPO Pattern: Input → Process → Output

Most beginner programs follow this pattern:

- **Input:** read values from the user (keyboard).
- **Process:** compute something using variables and operators.
- **Output:** show the result using `print()`.

```
name = input("Enter your name: ")
print("Hello,", name)
```

Important: `input()` always returns a `str`. So if you read numbers, you must convert:

```
a = input("Enter a number: ")
b = input("Enter another number: ")
total = int(a) + int(b)
print("Sum =", total)
```

If you forget `int(...)` here, then `"2" + "3"` becomes `"23"`, because string concatenation is used.

3 Demo Walkthrough: Mini Calculator

File: `demo/mini_calculator.py`

Goal

Read two numbers and print their sum, difference, product, and quotient. Also handle division by zero safely.

How to run

From the lecture folder:

```
python demo/mini_calculator.py
```

Key idea

Use `float(...)` to allow decimal inputs. Use an `if` condition to avoid dividing by zero.

4 Interactive Checkpoints (with Solutions)

Checkpoint 1 Solution

Question: When is interactive mode (REPL) better than a script?

Answer (examples):

- When you want to quickly test a small expression (e.g., check operator precedence).
- When you want to explore a library function without writing a full program.

Checkpoint 2 Solution

Question: What is the type of `input()`?

Answer: `input()` returns a `str`. You can verify using:

```
x = input("Enter something: ")
print(type(x)) # <class 'str'>
```

5 Practice Exercises (with Solutions)

Try these after the lecture. Solutions are provided so you can self-check.

Exercise 1: Sum of Two Integers

Task: Read two integers and print their sum.

Solution:

```
a = int(input("Enter a: "))
b = int(input("Enter b: "))
print("Sum =", a + b)
```

Exercise 2: Area of a Circle

Task: Read radius r and print area using πr^2 (use $\pi = 3.14159$).

Solution:

```
pi = 3.14159
r = float(input("Enter radius: "))
area = pi * r * r
print("Area =", area)
```

Exercise 3: Print a Clean Output Line

Task: Print Name: <name>, Age: <age> in one line.

Solution:

```
name = input("Name: ")
age = int(input("Age: "))
print("Name:", name + ",", "Age:", age)
```

6 Exit Question (with Solution)

Question: Write the command to run `task.py` from terminal.

Solution: `python task.py`

Appendix: Slide Deck Content (Reference)

The material below is a reference copy of the slide deck content. Exercise solutions are explained in the main notes where applicable.

Title Slide

Quick Links

[Core Concepts](#) [Demo](#) [Interactive](#) [Summary](#)

Agenda

- Overview
- Core Concepts
- Demo
- Interactive
- Summary

Learning Outcomes

- Explain **interactive (REPL)** vs **scripting** mode
- Run Python from terminal/IDLE/IDE
- Write a first program using `print()` and `input()`
- Follow the basic Input → Process → Output pattern

Why Python?

- Simple, readable syntax (good for beginners)
- Huge ecosystem (web, data, AI, automation)
- Cross-platform (Windows, Linux, macOS)
- Great for rapid prototyping and scripting

Two Ways to Run Python

- **Interactive mode (REPL)**: run one line at a time
- **Scripting mode**: save code in a `.py` file and run it

```
>>> 2 + 3
5
```

Interactive Mode (REPL)

Best for:

- quick calculations and testing small ideas
- learning syntax (instant feedback)
- checking library functions

```
>>> name = "UPES"  
>>> name.lower()  
'upes'
```

Scripting Mode (.py files)

Best for:

- programs you want to **save, reuse, and share**
- assignments, labs, mini-projects

Typical run command:

```
python my_program.py
```

First Program: Input → Process → Output

```
name = input("Enter your name: ")  
print("Hello,", name)
```

- `input()` always returns a `str`
- Use type conversion when you need numbers

Type Conversion (Quick Example)

```
a = input("Enter a number: ")  
b = input("Enter another number: ")  
total = int(a) + int(b)  
print("Sum =", total)
```

- Without `int()`, `"2" + "3"` becomes `"23"`

Demo: Mini Calculator

- File: `demo/mini_calculator.py`
- Reads two numbers and prints `+`, `-`, `*`, `/`
- Shows input conversion and division-by-zero handling

Demo Snippet

```
a = float(input("Enter a: "))
b = float(input("Enter b: "))
print("a + b =", a + b)
print("a - b =", a - b)
print("a * b =", a * b)
if b != 0:
    print("a / b =", a / b)
else:
    print("Division by zero is not allowed.")
```

Checkpoint 1

Question: When is interactive mode (REPL) better than a script?

Your answer: write 2 use-cases.

Checkpoint 2

Question: What is the type of `input()`?

```
x = input("Enter something: ")
print(type(x))
```

Think-Pair-Share

In pairs, discuss:

- Where can Python help you in daily life or engineering?
- Give 3 examples (automation / data / web / etc.).

Key Takeaways

- REPL is for quick exploration; scripts are for reusable programs
- `input()` returns a string; convert to `int/float` when needed
- Most programs follow Input → Process → Output

Exit Question

Write the command to run a Python script named `task.py` from terminal.